 Name ______________________________

CS 2614
Second Midterm Exam
Spring 2025

Name: ______Solutions______

Student Number: ______________________

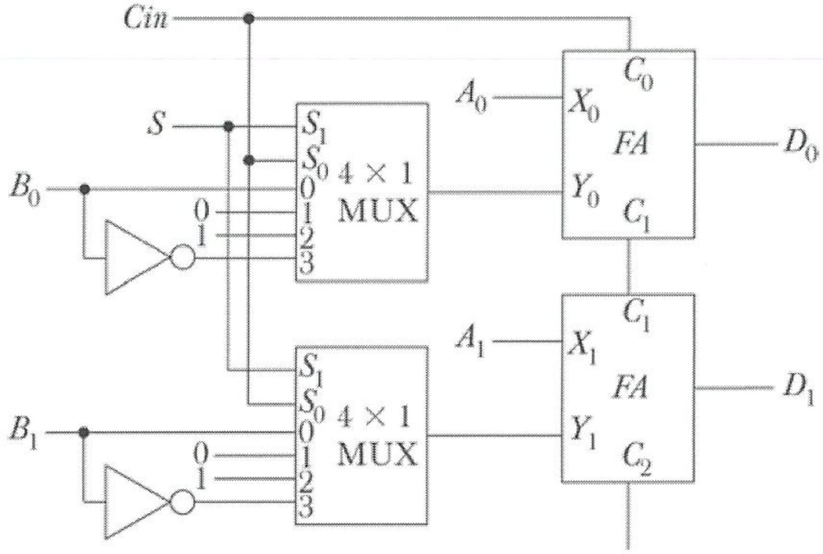
1. Place all bookbags and other materials in the back of classroom.
2. You may NOT use a calculator in any way or have it at your desk.
3. No references are allowed.
4. State all assumptions and show your work.
5. Place your answer in the spaces provided.
6. Answer all questions

Problem	% credit
1 (15%)	
2 (20%)	
3 (15%)	
4 (15%)	
5 (15%)	
6 (15%)	
7 (10%)	
Total	

Name ___________________________________

Question 1 [15 marks]

A and *B* are two 2-bit numbers. The arithmetic circuit shown below implements four arithmetic functions based on the two select input lines, *S* and C_{in}. In the truth table below, specify the arithmetic functions that are performed for various combinations of *S* and C_{in}

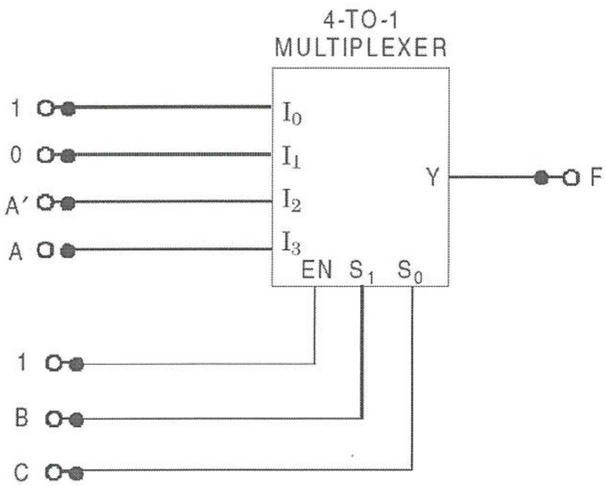


S	C_{in}	Arithmetic Function performed
0	0	A + B (Addition)
0	1	A + 1 (Increment)
1	0	A − 1 (Decrement)
1	1	A − B (Subtraction)

Name ________________________________

Question 2 [20 marks]

Draw the truth table implemented by the following hardware. Use the second row of the table to specify the input(s) and output(s) variables of the circuit. Use as many columns as needed in the table shown for the input and output variables.



Input variable(s)				Output variable(s)	
A	B	C		F	
0	0	0		1	
0	0	1		0	
0	1	0		1	
0	1	1		0	
1	0	0		1	
1	0	1		0	
1	1	0		0	
1	1	1		1	

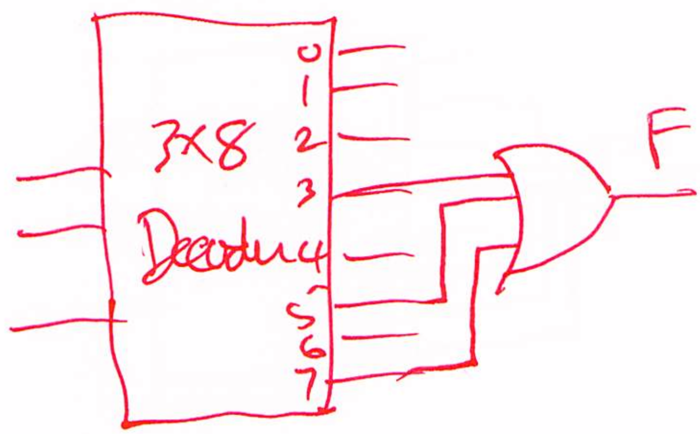
CS 2614

Spring 2025 Name ___________________________________

Question 3 [15 marks]

Draw the truth table for the function F (A,B,C) = BC + AC and then implement the function using a decoder and gates.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1



Question 4 [15 marks]

What size of ROM (in number of bits) is required to implement a combinational circuit which accepts a 2-bit number and generates a binary number equal to the square of the input number?

	Input	Output	
(0)	0 0	0 0 0 0	(0)
(1)	0 1	0 0 0 1	(1)
(2)	1 0	0 1 0 0	(4)
(3)	1 1	1 0 0 1	(9)

Size = 4x4

= 16

Ans (number of bits) = 16

CS 2614
Spring 2025 Name ________________________________

Question 5 [15 marks]

What is wrong with the following register transfer statements?

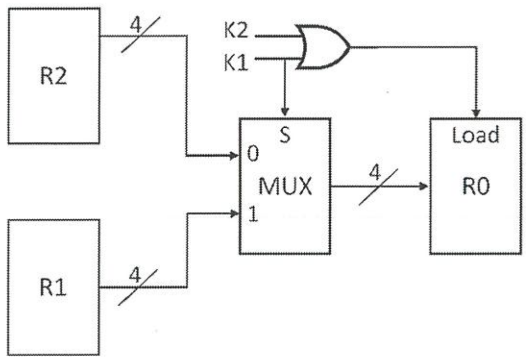
(a) xT: AR ← AR', AR ← AR + 1

AR is loaded simultaneously

(b) R1 ← R2, R1 ← R3

R1 is loaded concurrently

Question 6 [15 marks]
Write the Register Transfer Statement(s) that is/are implemented by the following hardware. R0, R1 and R2 are 4-bit registers.



K1: R0 ← R1

K1'K2: R0 ← R2

Ans=

Question 7 [10 marks]
Register A holds the 8-bit binary number 1101 1101. An XOR operation will be performed between Register A and Register B to change the value in Register A to 1101 0010. Determine the content of Register B to achieve the desired result.

A = 1101 1101
B = 0000 1111

A → 1101 0010

Ans = 0000 1111